

Member ID: _____

Time: _____

Rank: _____



SQL DATABASE FUNDAMENTALS (345) NATIONAL 2025

Multiple Choice:

50 @ 2 points each

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. What is the purpose of achieving Boyce-Codd Normal Form (BCNF) in database normalization?
 - A. To optimize data storage space
 - B. To reduce redundancy and dependency
 - C. To eliminate all duplicate records from the database
 - D. To ensure that each non-key attribute is fully functionally dependent on the primary key
2. Which of the following is NOT a type of database constraint?
 - A. Primary key constraint
 - B. Foreign key constraint
 - C. Null constraint
 - D. Dynamic constraint
3. How are OLTP systems typically scaled to handle increasing transaction loads?
 - A. By adding more powerful hardware resources
 - B. By partitioning data across multiple servers
 - C. By increasing the size of individual transactions
 - D. By reducing the number of concurrent users
4. Which database model is commonly used in OLTP systems?
 - A. Relational
 - B. NoSQL
 - C. Data warehousing
 - D. Hierarchical
5. What is the primary purpose of DDL in database management systems?
 - A. To manipulate data stored in the database
 - B. To define and modify the structure of the database schema
 - C. To generate complex analytical reports
 - D. To optimize database performance
6. What is the process of decomposing a relation into smaller, well-structured relations called?
 - A. Denormalization
 - B. Normalization
 - C. Decomposition
 - D. Aggregation

7. Which normal form allows for a composite key and eliminates partial dependencies?
- A. First Normal Form (1NF)
 - B. Second Normal Form (2NF)
 - C. Third Normal Form (3NF)
 - D. Boyce-Codd Normal Form (BCNF)

Consider the following tables for question 8.

Today's court bookings

Court	Start time	End time	Rate type
1	09:30	10:30	SAVER
1	11:00	12:00	SAVER
1	14:00	15:30	STANDARD
2	10:00	11:30	PREMIUM-B
2	11:30	13:30	PREMIUM-B
2	15:00	16:30	PREMIUM-A

8. What level of normalization does the following table meet?
- A. First Normal Form (1NF)
 - B. Second Normal Form (2NF)
 - C. Third Normal Form (3NF)
 - D. Boyce-Codd Normal Form (BCNF)
9. What does the term "atomicity" mean in the context of database normalization?
- A. Each transaction has been committed
 - B. Ensuring that each transaction is treated as a single unit
 - C. Ensures concurrent execution of transactions leaves the database in the same state
 - D. Ensures any data written to the database must be valid according to defined rules
10. In which isolation level does a transaction acquire range locks on selected rows and holds them until the end of the transaction?
- A. READ UNCOMMITTED
 - B. READ COMMITTED
 - C. REPEATABLE READ
 - D. SERIALIZABLE
11. Which of the following phenomena is associated with a dirty read?
- A. Uncommitted dependency
 - B. Lost update
 - C. Phantom read
 - D. Inconsistent retrieval
12. What is a nested stored procedure?
- A. A stored procedure that contains multiple levels of control flow statements
 - B. A stored procedure that is called by another stored procedure
 - C. A stored procedure that is created within another stored procedure
 - D. A stored procedure that uses recursive logic to process hierarchical data

13. What is a cursor in the context of stored procedures?
 - A. A temporary table used to store intermediate results
 - B. A control structure used to iterate over a result set row by row
 - C. A parameter used to pass data between stored procedures
 - D. A mechanism for exception handling within a stored procedure
14. What does CAP theorem state about distributed systems, including NoSQL databases?
 - A. They can have only one of the following: consistency, availability, or partition tolerance
 - B. They must sacrifice consistency in favor of availability and partition tolerance
 - C. They must sacrifice availability in favor of consistency and partition tolerance
 - D. They must sacrifice partition tolerance in favor of consistency and availability
15. What is the main advantage of using a document-based NoSQL database?
 - A. Strong schema enforcement
 - B. Support for complex joins and transactions
 - C. Flexible schema design and easy scalability
 - D. Strict ACID properties for data consistency
16. Which aspect of DBaaS allows for seamless scaling of database resources?
 - A. Automated backups
 - B. Elastic provisioning
 - C. Manual sharding
 - D. Fixed instance sizes
17. What does DBaaS abstract away from users?
 - A. Database security
 - B. Database performance tuning
 - C. Database hardware and software provisioning
 - D. Database schema design
18. How does polyglot persistence relate to the microservices architecture?
 - A. Polyglot persistence is a prerequisite for implementing microservices
 - B. Microservices and polyglot persistence are unrelated concepts
 - C. Microservices often use polyglot persistence to manage data diversity
 - D. Polyglot persistence simplifies the deployment of microservices
19. What role does data integration play in polyglot persistence?
 - A. Data integration is not necessary in polyglot persistence environments
 - B. Data integration ensures consistency across multiple databases
 - C. Data integration facilitates communication between different database systems
 - D. Data integration increases complexity and should be avoided

20. Which database optimization strategy involves partitioning large tables into smaller, more manageable segments?
- A. Database sharding
 - B. Database clustering
 - C. Database replication
 - D. Database normalization
21. What distinguishes a data lake from a traditional data warehouse?
- A. Data lakes store only structured data, while data warehouses store both structured and unstructured data.
 - B. Data lakes are designed for real-time processing, while data warehouses are optimized for batch processing.
 - C. Data lakes allow for schema-on-read, enabling flexibility in data processing, while data warehouses enforce schema-on-write.
 - D. Data lakes are typically deployed on-premises, while data warehouses are typically deployed in the cloud.
22. What is data ingestion in the context of data lakes?
- A. The process of exporting data from a data lake to other systems
 - B. The process of transforming unstructured data into structured data
 - C. The process of loading data into a data lake from various sources
 - D. The process of querying data stored in a data lake
23. Which of the following is NOT a common use case for data lakes?
- A. Business intelligence and analytics
 - B. Machine learning and data science
 - C. Log and clickstream analysis
 - D. Real-time transaction processing
24. What is RBAC in database security?
- A. The process of encrypting data to prevent unauthorized access
 - B. The process of restricting access to database objects based on user roles and privileges
 - C. The process of monitoring database activity for suspicious behavior
 - D. The process of restricting access to certain IP add
25. What is data masking in the context of database security?
- A. The process of obfuscating sensitive data in non-production environments
 - B. The process of backing up data to prevent data loss in case of a disaster
 - C. The process of monitoring and recording database activity for compliance purposes
 - D. The process of encrypting data during transmission over a network

26. How does BASE address the challenge of network partitions in distributed systems?
- A. By prioritizing consistency over availability
 - B. By ensuring that all replicas of data are consistent at all times
 - C. By sacrificing immediate consistency for increased availability and partition tolerance
 - D. By preventing failures from affecting the overall system operation
27. In data lineage, what does a "data flow" represent?
- A. The movement of data between different database tables
 - B. The encryption of data during transmission
 - C. The sequence of operations performed on data within a database
 - D. The path data takes as it moves through systems and processes
28. In ER modeling, what does cardinality represent?
- A. The number of attributes in an entity
 - B. The degree of a relationship between entities
 - C. The type of data stored in a database table
 - D. The size of a database schema
29. What is a subtype in an ER model?
- A. An entity that inherits attributes and relationships from another entity
 - B. An entity with a many-to-many relationship
 - C. An entity with a composite primary key
 - D. An entity that represents a specialized version of another entity
30. What does the term "isolation" mean in the context of database normalization?
- A. Ensuring that each transaction is treated as a single unit
 - B. Ensures any data written to the database must be valid according to defined rule
 - C. Each transaction has been committed
 - D. Ensures concurrent execution of transactions leaves the database in the same state
31. What is aggregation in ER modeling?
- A. A process for combining multiple database tables into a single table
 - B. A process for decomposing a database schema into smaller, more manageable components
 - C. A process for modeling relationships between entities at different levels of abstraction
 - D. A process for optimizing database performance by reducing redundancy
32. In the context of database migration, what does the term "data drift" refer to?
- A. Inconsistencies between data in the old and new databases
 - B. Unauthorized access to sensitive data during migration
 - C. Migration of data across geographical regions
 - D. Replication lag in distributed databases

33. Which of the following is NOT a characteristic of big data?
- A. Veracity
 - B. Variability
 - C. Value
 - D. Viscosity
34. In which scenarios are materialized views most beneficial?
- A. When real-time data access is critical
 - B. When storage space is limited
 - C. When the view is accessed frequently
 - D. When data security is the primary concern
35. Which statement best describes the concept of privilege escalation in DCL?
- A. Granting additional privileges to a user or role
 - B. Removing existing privileges from a user or role
 - C. Allowing users to access data they are not authorized to access
 - D. Modifying database schema and structure
36. Which of the following indexing strategies is commonly used for columns with high cardinality and range queries?
- A. B-tree index
 - B. Bitmap index
 - C. Hash index
 - D. Full-text index
37. Which of the following algorithms are commonly used for association rule mining?
- A. K-means clustering
 - B. Apriori
 - C. Decision trees
 - D. Linear regression
38. What is the curse of dimensionality in data mining?
- A. A phenomenon where the accuracy of a classifier decreases as the number of features increases
 - B. An issue that arises when attempting to replicate data on demand
 - C. A problem that occurs when imputing missing values in a dataset
 - D. An unexpected result of reporting data from different sources simultaneously
39. What is the main challenge of association rule mining?
- A. Handling missing values in the dataset
 - B. Dealing with high-dimensional data
 - C. Extracting meaningful patterns from large datasets
 - D. Interpreting the discovered rules and making them actionable

40. Which of the following statements about column-family stores is true?
- A. They are optimized for OLTP workloads
 - B. They store all columns of a row contiguously on disk
 - C. They are not suitable for analytical queries and reporting
 - D. They use a schema-less design
41. In a distributed database system, what is a key challenge?
- A. Ensuring data consistency across multiple nodes
 - B. Minimizing storage space requirements
 - C. Simplifying data access for end-users
 - D. Achieving high availability for the database
42. Which of the following is an advantage of using a columnar database over a row-based database for analytical workloads?
- A. Faster transaction processing
 - B. Lower storage requirements
 - C. Higher compression ratios
 - D. Better support for OLTP workloads
43. Which of the following concurrency control mechanisms ensures serializability by locking data items that are accessed by transactions?
- A. Two-Phase Locking (2PL)
 - B. Concurrency Control (OCC)
 - C. Multi-Version Concurrency Control (MVCC)
 - D. Snapshot Isolation (SSI)

Consider the following tables for questions 44-46: *EmployeeID* is auto-increment, unique, and primary key;

Table Name: **Employee**

EmployeeID	FirstName	LastName	DepartmentID
1	TheMighty	Thor	101
2	Wonder	Woman	102
3	Spider	Man	101
4	Scarlett	Witch	104

Table Name: **Departments**

ID	DepartmentName
101	Marketing
102	Sales
103	Finance
104	Human Resources

44. What would be the result of the following query:

SELECT 123 + "789"+ 456 from Employee

- A. Error
- B. 579
- C. 1368 4 times
- D. "123789456"

45. What DDL command would you use to add a Foreign key constraints to the Employee table?

- A. ALTER TABLE
- B. ADD FOREIGN KEY CONSTRAINT
- C. MODIFY TABLE
- D. ADD CONSTRAINT

46. How would you get a count of how many employees are in each department?
- A. `SELECT DepartmentName, COUNT(EmployeeID) FROM Employees
INNER JOIN Departments ON
Employees.EmployeeID = Departments.ID
GROUP BY DepartmentName`
 - B. `SELECT D.DepartmentName, COUNT(E.EmployeeID) FROM Employees E
LEFT JOIN Departments D ON
D.EmployeeID = E.ID
GROUP BY D.DepartmentName`
 - C. `SELECT DepartmentName, COUNT(EmployeeID) FROM Departments Dept
LEFT JOIN Employees Emp ON
EmployeeID = Dept.ID
GROUP BY DepartmentName`
 - D. `SELECT DepartmentName, COUNT(EmployeeID) FROM Employees
CROSS JOIN Departments ON
Employees.EmployeeID = Departments.ID
GROUP BY DepartmentName`
47. What is the purpose of the EXPLAIN statement in SQL?
- A. To execute a query and retrieve the result set
 - B. To display the execution plan of a query
 - C. To provide a summary of database statistics
 - D. To analyze the data distribution within tables
48. In a star schema data warehouse design, what is the role of the fact table?
- A. To store metadata about the database objects
 - B. To store descriptive attributes about the business entities
 - C. To store hierarchical relationships between data entities
 - D. To store transactional data or business facts
49. Which of the following is a characteristic of a NewSQL database?
- A. It is designed for vertical and not horizontal scaling
 - B. It is optimized for ultra-heavy workloads
 - C. It supports only eventual consistency
 - D. It functionally supports OLTP databases
50. What does the term "Consistency" mean in the context of database normalization?
- A. Ensures any data written to the database must be valid according to defined rules
 - B. Each transaction has been committed
 - C. Ensures concurrent execution of transactions leaves the database in the same state
 - D. Ensuring that each transaction is treated as a single unit